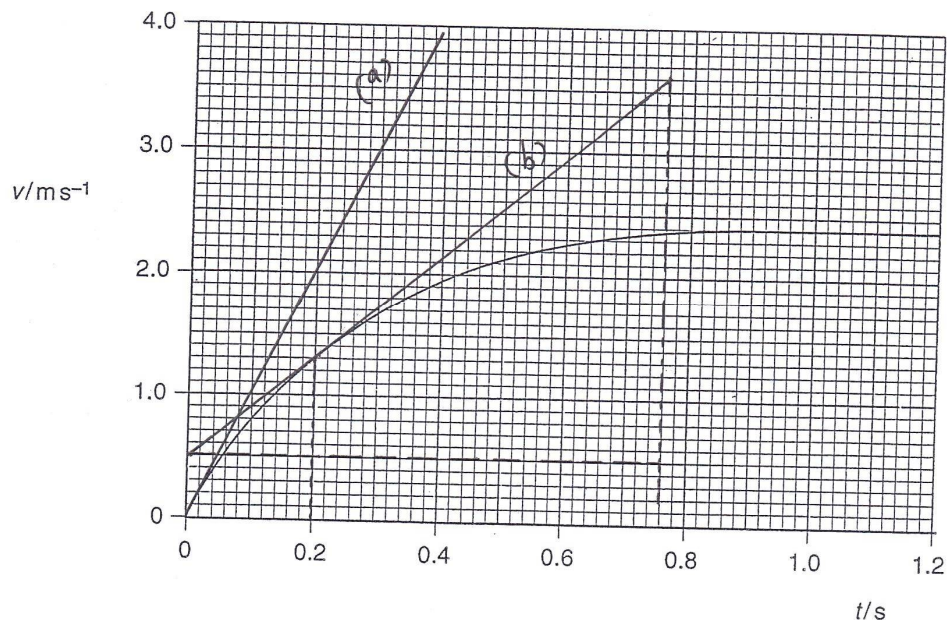


Section A

1 (a) & (b) answer on the Fig. 1.1



(b) acceleration = gradient of the tangent at $t = 0.20$ s.

$$\text{gradient} = \frac{3.1}{0.76} = 4.08 \text{ m s}^{-2}$$

(c) (i) The maximum resistive force acting on the ball is when the ball reaches terminal velocity. Hence the maximum resistive force acting on the ball is equal to the weight of the ball, $F = (0.015 \times 9.81) = 0.147 \text{ N}$

(ii) at $t = 0.20$ s, $mg - F = ma$

$$F = mg - ma$$

$$F = 0.147 - (0.015)(4.08) = 0.0858 \text{ N}$$

(d) If the magnitude of air resistance is proportional to the speed of the ball, then the ratio of F_R/v must be a constant.

$$\text{from (c)(i) and Fig 1.1, } \frac{F_R}{v} = \frac{0.147}{2.4} = 0.0613$$

$$\text{from (c)(ii) and Fig. 1.1, } \frac{F_R}{v} = \frac{0.083}{1.3} = 0.0638$$

the two ratio differs by 0.0026 (approximately 4.2%).

As the difference is small, the two ratios can be assumed to be the same. The air resistance is proportional to the speed of the ball.